

US00PP33738P2

(12) **United States Plant Patent**
Stewart et al.

(10) **Patent No.:** **US PP33,738 P2**
(45) **Date of Patent:** **Dec. 14, 2021**

- (54) **STRAWBERRY PLANT NAMED ‘DRISSTRAW EIGHTYSEVEN’**
- (50) Latin Name: *Fragara x ananassa*
Varietal Denomination: **DrisStrawEightySeven**
- (71) Applicant: **Driscoll’s, Inc.**, Watsonville, CA (US)
- (72) Inventors: **Philip J. Stewart**, Watsonville, CA (US); **Esther J. Kibbe**, Watsonville, CA (US); **Raymond L. Jacobs, III**, Watsonville, CA (US); **Mary M. Calkins**, Watsonville, CA (US)
- (73) Assignee: **Driscoll’s, Inc.**, Watsonville, CA (US)
- (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.
- (21) Appl. No.: **17/177,032**
- (22) Filed: **Feb. 16, 2021**
- (51) **Int. Cl.**
A01H 5/08 (2018.01)
A01H 6/74 (2018.01)
- (52) **U.S. Cl.**
USPC **Plt./209**
- (58) **Field of Classification Search**
USPC Plt./156, 208, 209
See application file for complete search history.
- (56) **References Cited**

U.S. PATENT DOCUMENTS

PP1,745 P 8/1958 Lang
PP3,981 P 11/1976 Bringhurst et al.
PP4,487 P 11/1979 Bringhurst et al.
PP4,538 P 5/1980 Bringhurst et al.
PP5,262 P 7/1984 Voth et al.
PP5,265 P 7/1984 Voth et al.
PP5,266 P 7/1984 Bringhurst et al.
PP5,300 P 10/1984 Johnson, Jr. et al.
PP5,840 P 12/1986 Johnson, Jr. et al.
PP6,191 P 5/1988 Johnson, Jr. et al.
PP6,231 P 7/1988 Johnson, Jr. et al.
PP6,578 P 1/1989 Voth et al.
PP6,579 P 1/1989 Bringhurst et al.
PP7,024 P 9/1989 Johnson, Jr. et al.
PP7,172 P 2/1990 Voth et al.
PP7,522 P 5/1991 Johnson, Jr. et al.
PP7,614 P 8/1991 Bringhurst et al.
PP7,615 P 8/1991 Bringhurst et al.
PP8,086 P 1/1993 Nelson et al.
PP8,205 P 4/1993 Nelson et al.
PP8,649 P 3/1994 Sjulín et al.
PP8,661 P 3/1994 Bringhurst et al.
PP8,708 P 5/1994 Voth et al.
PP8,745 P 5/1994 Sjulín et al.
PP9,130 P 5/1995 Sjulín et al.
PP9,909 P 6/1997 Ackerman et al.
PP10,221 P 2/1998 Sjulín et al.
PP10,534 P 8/1998 Sjulín et al.
PP10,642 P 10/1998 Amorao et al.
PP11,035 P 8/1999 Mowrey et al.
PP11,277 P 3/2000 Gilford et al.
PP11,279 P 3/2000 Gilford et al.
PP11,522 P 9/2000 Amorao et al.

PP11,548 P 10/2000 Amorao et al.
PP11,554 P 10/2000 Sjulín et al.
PP11,639 P 11/2000 Mowrey et al.
PP12,186 P2 11/2001 Gilford et al.
PP12,436 P2 3/2002 Amorao et al.
PP12,577 P2 4/2002 Amorao et al.
PP12,817 P2 7/2002 Gilford et al.
PP12,899 P2 9/2002 Mowrey et al.
PP13,386 P2 12/2002 Mowrey et al.
PP13,469 P3 1/2003 Larson et al.
PP14,005 P3 7/2003 Amorao et al.
PP14,062 P3 8/2003 Amorao et al.
PP14,109 P3 8/2003 Gilford et al.
PP14,771 P3 5/2004 Amorao et al.
PP15,145 P2 9/2004 Mowrey et al.
PP15,308 P2 11/2004 Sjulín et al.
PP15,375 P2 11/2004 Mowrey et al.
PP15,435 P2 12/2004 Sjulín et al.
PP15,596 P2 3/2005 Amorao et al.
PP15,731 P2 4/2005 Amorao et al.
PP15,752 P2 5/2005 Gilford et al.
PP16,070 P2 10/2005 Gilford et al.
PP16,238 P2 2/2006 Amorao et al.
PP16,241 P2 2/2006 Mowrey et al.
PP16,298 P2 2/2006 Gilford et al.
PP16,299 P2 2/2006 Gilford et al.
PP16,475 P2 4/2006 Gilford et al.
PP16,558 P3 5/2006 Lopez
PP18,000 P2 9/2007 Meulenbroek
PP18,040 P3 9/2007 Mowrey et al.
PP18,041 P3 9/2007 Gilford
PP18,458 P2 1/2008 Ferguson et al.
PP18,575 P3 3/2008 Amorao et al.
PP18,878 P2 6/2008 Mowrey et al.
PP19,240 P2 9/2008 Gilford et al.
PP19,673 P3 2/2009 Ferguson et al.
PP19,767 P2 2/2009 Shaw et al.
PP20,248 P3 9/2009 Rogers et al.
PP20,363 P2 9/2009 Chandler
PP20,701 P2 2/2010 Gilford et al.
PP20,731 P2 2/2010 Mowrey et al.
PP20,733 P2 2/2010 Mowrey et al.
PP20,735 P2 2/2010 Ferguson
PP20,775 P2 2/2010 Mowrey et al.
PP20,922 P2 4/2010 Gilford et al.
PP21,538 P2 11/2010 Gilford et al.
PP21,559 P2 12/2010 Ferguson et al.
PP21,762 P2 3/2011 Stewart et al.
PP22,040 P3 7/2011 Stewart et al.
PP22,218 P2 11/2011 Ferguson et al.
PP22,247 P2 11/2011 Ferguson
PP23,107 P2 10/2012 Ferguson et al.
PP23,148 P2 10/2012 Gilford et al.
PP23,377 P2 2/2013 Ferguson et al.
PP23,378 P2 2/2013 Pullen et al.
PP23,382 P2 2/2013 Ferguson et al.
PP23,383 P2 2/2013 Ferguson et al.

(Continued)

Primary Examiner — Susan McCormick Ewoldt
Assistant Examiner — Karen M Redden
(74) Attorney, Agent, or Firm — Morrison & Foerster LLP

(57) **ABSTRACT**

A new and distinct variety of strawberry plant named ‘DrisStrawEightySeven’, particularly selected for its earliness of production, as well as the shape, appearance, flavor, and shelf life of its fruit, is disclosed.

(56)

References Cited

U.S. PATENT DOCUMENTS

PP23,400	P2	2/2013	Ferguson et al.	PP29,729	P2	10/2018	Kibbe et al.
PP23,401	P2	2/2013	Pullen et al.	PP29,730	P2	10/2018	Kibbe et al.
PP23,459	P2	3/2013	Stewart et al.	PP29,731	P2	10/2018	Ferguson et al.
PP23,506	P3	4/2013	Ferguson et al.	PP29,747	P2	10/2018	Vitten et al.
PP23,517	P3	4/2013	Ferguson et al.	PP29,748	P2	10/2018	Vitten et al.
PP24,096	P3	12/2013	Fear et al.	PP29,749	P2	10/2018	Stewart et al.
PP24,317	P3	3/2014	Ferguson et al.	PP30,775	P2	8/2019	Carrillo Mendoza et al.
PP24,333	P3	3/2014	Vitten et al.	PP30,789	P2	8/2019	Ferguson et al.
PP24,395	P3	4/2014	Vitten et al.	PP30,818	P2	8/2019	Pakozdi et al.
PP24,533	P3	6/2014	Ferguson et al.	PP30,851	P2	8/2019	Fear et al.
PP24,745	P2	8/2014	Vitten et al.	PP30,936	P2	10/2019	Stewart et al.
PP25,408	P3	4/2015	Vitten et al.	PP31,083	P2	11/2019	Carrillo Mendoza et al.
PP25,437	P3	4/2015	Vitten et al.	PP31,233	P2	12/2019	Pakozdi et al.
PP25,698	P3	7/2015	Ferguson et al.	PP31,527	P3	3/2020	Carrillo Mendoza et al.
PP25,699	P3	7/2015	Stewart et al.	PP31,655	P2	4/2020	Carrillo Mendoza et al.
PP25,747	P3	7/2015	Kibbe et al.	PP31,703	P2	4/2020	Stewart et al.
PP25,866	P3	9/2015	Ferguson et al.	PP31,827	P2	6/2020	Ferguson et al.
PP26,800	P3	6/2016	Stewart et al.	PP31,896	P2	6/2020	Pakozdi et al.
PP26,801	P3	6/2016	Stewart et al.	PP31,935	P2	7/2020	Pakozdi et al.
PP26,802	P3	6/2016	Rodriguez Alcazar et al.	PP32,079	P2	8/2020	Carrillo Mendoza et al.
PP27,442	P2	12/2016	Kibbe et al.	PP32,080	P2	8/2020	Ferguson et al.
PP27,645	P3	2/2017	Vitten et al.	PP32,271	P2	10/2020	Ferguson et al.
PP27,682	P3	2/2017	Kibbe et al.	PP32,305	P3	10/2020	Pakozdi et al.
PP27,711	P3	2/2017	Vitten et al.	PP32,498	P2	11/2020	Carrillo Mendoza et al.
PP27,813	P3	3/2017	Ferguson et al.	PP32,499	P3	11/2020	Jacobs et al.
PP29,289	P3	5/2018	Vitten et al.	PP32,500	P3	11/2020	Stewart et al.
PP29,728	P2	10/2018	Stewart et al.	PP32,801	P2	2/2021	Ferguson et al.
				PP32,814	P2	2/2021	Stewart et al.
				PP32,824	P2	2/2021	Stewart et al.
				2003/0079263	P1	4/2003	Gilford et al.
				2013/0276182	P1	10/2013	Fear et al.

1

**STRAWBERRY PLANT NAMED
'DRISSTRAWEIGHTYSEVEN'**

Botanical classification: *Fragaria* x *ananassa*.

Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawEightySeven'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18th century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawEightySeven'.

Strawberry plant variety 'DrisStrawEightySeven' originated from a controlled cross between the female parent 'DrisStrawFiftyOne' (U.S. Plant Pat. No. 29,730) and the proprietary male parent '45AB129' (unpatented). Progeny plants from this cross, including 'DrisStrawEightySeven', were asexually propagated via stolons in Shasta County, Calif. in May of 2015. Strawberry plant variety 'DrisStrawEightySeven' was later specifically identified and selected in Hillsborough County, Fla. in January of 2016.

'DrisStrawEightySeven' was subsequently asexually propagated via stolons, and has undergone testing at test plots in Hillsborough County, Fla. for six years (2015 to 2021). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons and tissue culture.

'DrisStrawEightySeven' was particularly selected for its earliness of production, as well as the shape, appearance, flavor, and shelf life of its fruit.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs. The colors shown are as true as can be

2

reasonably obtained by conventional photographic procedures. Unless otherwise indicated, the photographs are of plants that are four months old.

FIG. 1 illustrates whole fruit of variety 'DrisStrawEightySeven'.

FIG. 2 illustrates longitudinal sections of fruit of variety 'DrisStrawEightySeven'.

FIG. 3 illustrates the upper surface (top row) and lower surface (bottom row) of flowers of variety 'DrisStrawEightySeven'.

FIG. 4 illustrates the lower surface (top) and upper surface (bottom) of leaves of variety 'DrisStrawEightySeven'.

FIG. 5 illustrates whole plants of variety 'DrisStrawEightySeven'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawEightySeven'. The data which define these characteristics is based on observations taken in Hillsborough County, Fla. from 2015 to 2021. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawEightySeven' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawEightySeven' was taken from plants that were four months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.) (2015 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary*, 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:

Species.—*Fragaria* x *ananassa*.

Common name.—Strawberry.

Denomination.—'DrisStrawEightySeven'.

Parentage:

Female parent.—'DrisStrawFiftyOne' (U.S. Plant Pat. No. 29,730).

Male parent.—Proprietary strawberry plant '45AB129' (unpatented).

Plant:

Height.—18.4 cm.

Diameter.—31.8 cm.

Height/width ratio.—0.58.

Number of crowns per plant.—1.4.

Growth habit.—Semi-upright.

Density of foliage.—Sparse.

Vigor.—Medium.

Stolon:

Number of daughter plants per square foot.—26.

Diameter at bract.—3.0 mm.

Color.—RHS 144B (Strong yellowish green).

Anthocyanin coloration.—Medium.

Anthocyanin color.—RHS 37B (Strong yellowish pink).

Density of pubescence.—Medium.

Fruiting truss:

Length from crown to base of terminal flower or fruit.—20.2 cm.

Diameter at base of truss.—0.56 cm.

Number of berries per fruiting truss.—4.2.

Attitude at first picking.—Prostrate.

3

- Color at base of truss.*—RHS 145B (Light yellowish green).
- Leaf:
- Number of leaflets.*—Three only.
- Color of leaf upper surface.*—RHS 137A (Moderate olive green). 5
- Color of leaf lower surface.*—RHS 145B (Light yellowish green).
- Blistering.*—Medium. 10
- Glossiness.*—Medium.
- Variation.*—Absent.
- Terminal leaflet.*—Length: 9.5 cm. Width: 9.2 cm. Length/width ratio: 1.03. Number of teeth per terminal leaflet: 22.7. Shape: Obicular. Shape of base: Obtuse. Shape of apex: Rounded. Margin: Serrate to crenate. Margin profile: Flat (level with the leaflet blade). Shape in cross section: Concave. 15
- Petiole.*—Length: 158 mm. Diameter: 3.8 mm. Color: RHS 145B (Light yellowish green). Pubescence: Dense. Attitude of hairs: Horizontal. Bract frequency (number present on each petiole): 0. 20
- Petiolule.*—Length: 10.8 mm. Diameter: 1.93 mm. Color: RHS 145B (Light yellowish green).
- Stipule.*—Length: 44.5 mm. Width: 8.6 mm. Pubescence: Medium. Color: RHS 150D (Light yellowish green). Anthocyanin coloration: Weak. Anthocyanin color: RHS 145A (Strong yellowish green). 25
- Inflorescence:
- Number of flowers per plant.*—3.9. 30
- Position of inflorescence in relation to foliage.*—Same level.
- Pedicel.*—Attitude of hairs: Upwards.
- Flower.*—Flower diameter (petal tip to petal tip on non-flattened flower): 34.7 mm. Arrangement of petals: Overlapping. Size of calyx in relation to corolla: Larger. Diameter of inner calyx relative to outer: Same size. Receptacle color: RHS 145B (Light yellowish green). Stamen: Present. 35
- Petal.*—Length: 14.8 mm. Width: 15.6 mm. Length/width ratio: 0.95. Number of petals per flower: 6.7. Color of upper side: RHS 155C (Greenish White). Color of lower side: RHS 155C (Greenish White). Shape: Obicular. Shape of apex: Rounded. Shape of base: Convex. Margin: Entire. 40
- Calyx.*—Diameter (sepal tip to sepal tip, measured on back of flower): 65.0 mm. 45
- Sepal.*—Length: 28.6 mm. Width: 10.6 mm. Number of sepals per flower: 13.4. Shape: Elliptical. Shape of apex: Truncate. Margin: Serrate.
- Anther.*—Color: RHS 154A (Vivid yellowish green).
- Flowering.*—Flowering interval: Late October to March.
- Fruit:
- Fruit size.*—Length: 53.2 mm. Width: 38.7 mm. Length/width ratio: 1.37. 55
- Fruit hollow.*—Length: 27.8 mm. Width: 8.7 mm. Length/width ratio: 3.19.
- Shape.*—Ovoid to cylindrical.
- Difference in shape of terminal and other fruits.*—None or very slight. 60
- Fruit color.*—RHS 43A (Deep red).
- Evenness of color.*—Slightly uneven.
- Glossiness.*—Medium.
- Evenness of surface.*—Even or very slightly uneven. 65
- Width of band without achenes.*—Narrow.
- Position of achenes.*—Below surface.

4

- Achenes.*—Number of achenes per fruit: 622. Weight: 0.0004119 g/achene. Color (sunward side of berry): RHS 178B (Dark reddish orange). Color (shaded side of berry): RHS 151C (Strong greenish yellow).
- Position of calyx attachment.*—Raised.
- Attitude of sepals.*—Downwards.
- Diameter of calyx in relation to diameter of fruit.*—Much larger.
- Adherence of calyx.*—Very strong.
- Firmness.*—Firm.
- Color of flesh (excluding core).*—RHS 31B (Strong reddish orange).
- Evenness of color of the flesh.*—Slightly uneven.
- Distribution of flesh color.*—Marginal and central.
- Color of core.*—RHS 27A (Light yellowish pink).
- Weight.*—26.8 g.
- Soluble solids (in ° brix).*—7.6.
- Titrateable acidity (as % citric acid).*—2.5.
- Fruiting.*—Harvest interval: Mid-November to March. Type of bearing: Partially remontant. Productivity: 0.528 kg to 0.741 kg of fruit per plant per season from seven-month-old plants when grown in Dover, Fla.
- Storage characteristics (post-harvest characteristics).*—Following harvest, fruit maintains its glossy appearance with minimal loss of fruit quality when stored for 7 days under refrigerated conditions that are standard for strawberry storage.
- Resistance to abiotic stress, pests, and diseases:
- Heat.*—Moderately resistant.
- Rain damage.*—Resistant.
- Two-spotted spider mite (tetranychus urticae).*—Moderately susceptible.
- Botrytis fruit rot (botrytis cinerea).*—Moderately susceptible.
- Powdery mildew (podosphaera macularis).*—Moderately susceptible.
- Anthracoze fruit rot (colletotrichum acutatum).*—Susceptible.
- Anthracoze crown rot (colletotrichum gloeosporioides).*—Moderately susceptible.
- Charcoal rot (macrohomina phaseolina).*—Moderately susceptible.
- Angular leaf spot (xanthomonas fragariae).*—Susceptible.

COMPARISON WITH PARENTAL AND
REFERENCE VARIETIES

- ‘DrisStrawEightySeven’ differs from the female parent ‘DrisStrawFiftyOne’ (U.S. Plant Pat. No. 29,730) in that ‘DrisStrawEightySeven’ is resistant to rain damage, whereas ‘DrisStrawFiftyOne’ is very susceptible to rain damage.
- ‘DrisStrawEightySeven’ differs from the male parent proprietary strawberry plant ‘45AB129’ (unpatented) in that ‘DrisStrawEightySeven’ does not produce creased fruit and all fruit has singular tips, whereas ‘45AB129’ produces some fruit that is creased and some fruit that has double tips.
- ‘DrisStrawEightySeven’ differs from the reference variety ‘DrisStrawTwentyFour’ (U.S. Plant Pat. No. 23,378) in that ‘DrisStrawEightySeven’ has a semi-upright growth habit, a convex petal shape of base, a narrow width of band without achenes on fruit, and a below surface position of achenes on fruit, whereas ‘DrisStrawTwentyFour’ has a spreading growth habit, a concavo-convex petal shape of base, a broad width of band without achenes on fruit, and an above surface position of achenes on fruit.

‘DrisStrawEightySeven’ differs from the reference variety ‘DrisStrawFortyNine’ (U.S. Plant Pat. No. 27,682), in that ‘DrisStrawEightySeven’ has medium stolon density of pubescence, a convex petal shape of base, an elliptical sepal shape, and a raised position of calyx attachment on fruit, 5 whereas ‘DrisStrawFortyNine’ has absent or very sparse stolon density of pubescence, a concavo-convex petal shape of base, an oblong sepal shape, and an inserted position of calyx attachment on fruit.

We claim:

10

1. A new and distinct variety of strawberry plant named ‘DrisStrawEightySeven’ as shown and described herein.

* * * * *

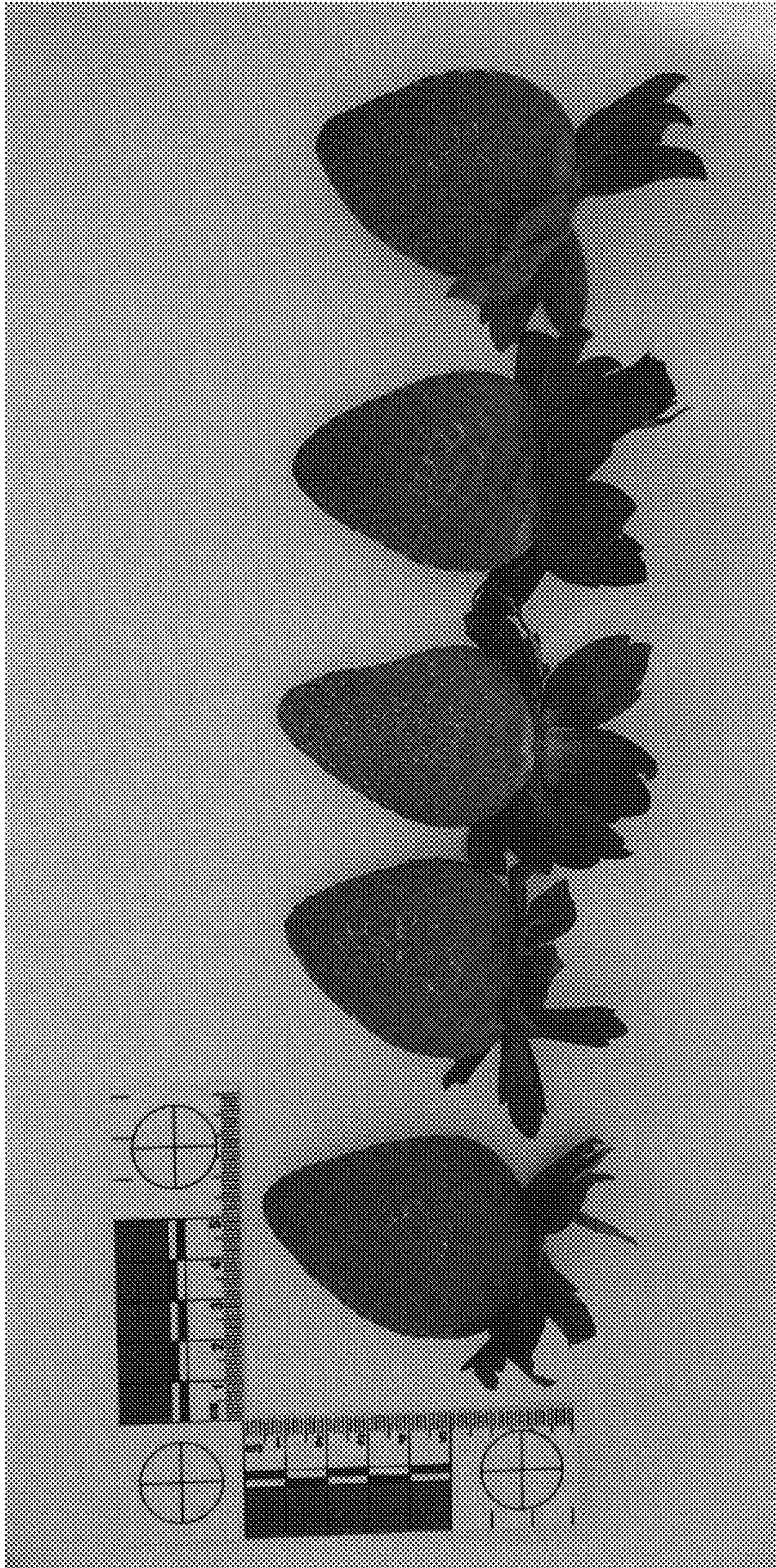


FIG. 1

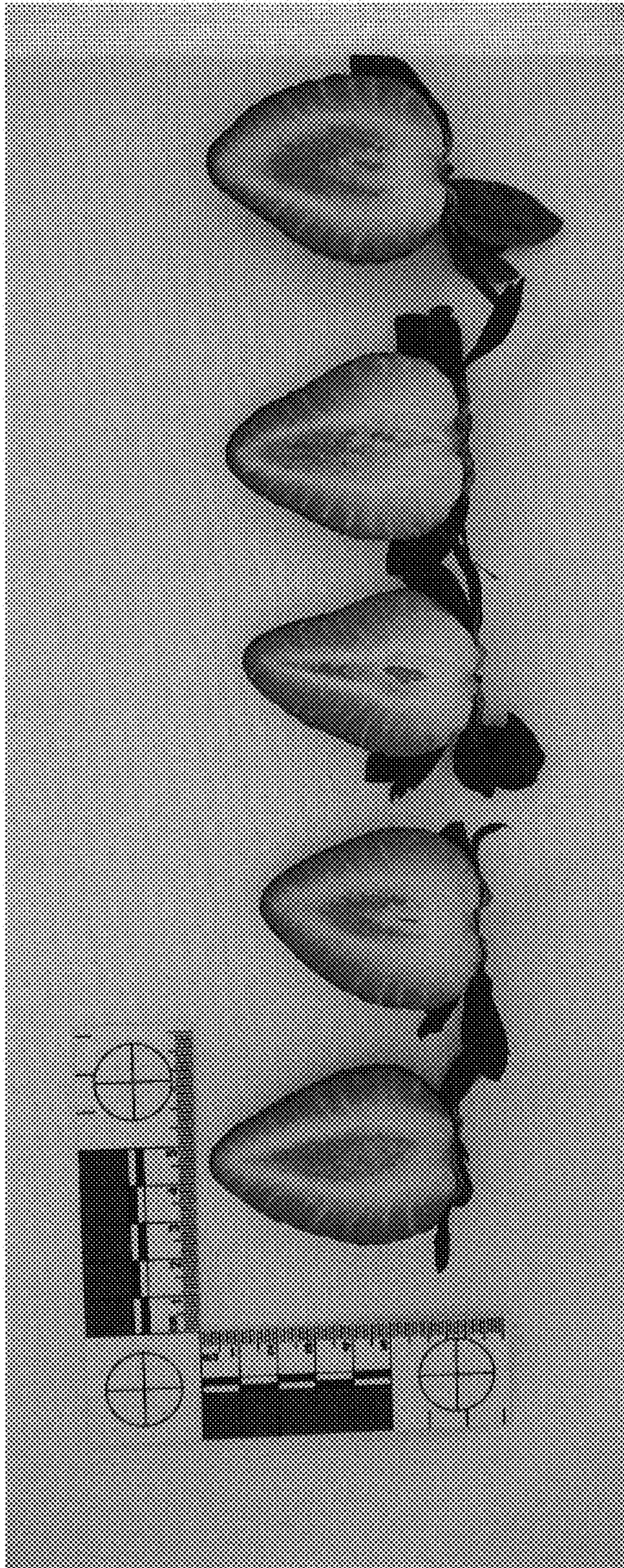


FIG. 2

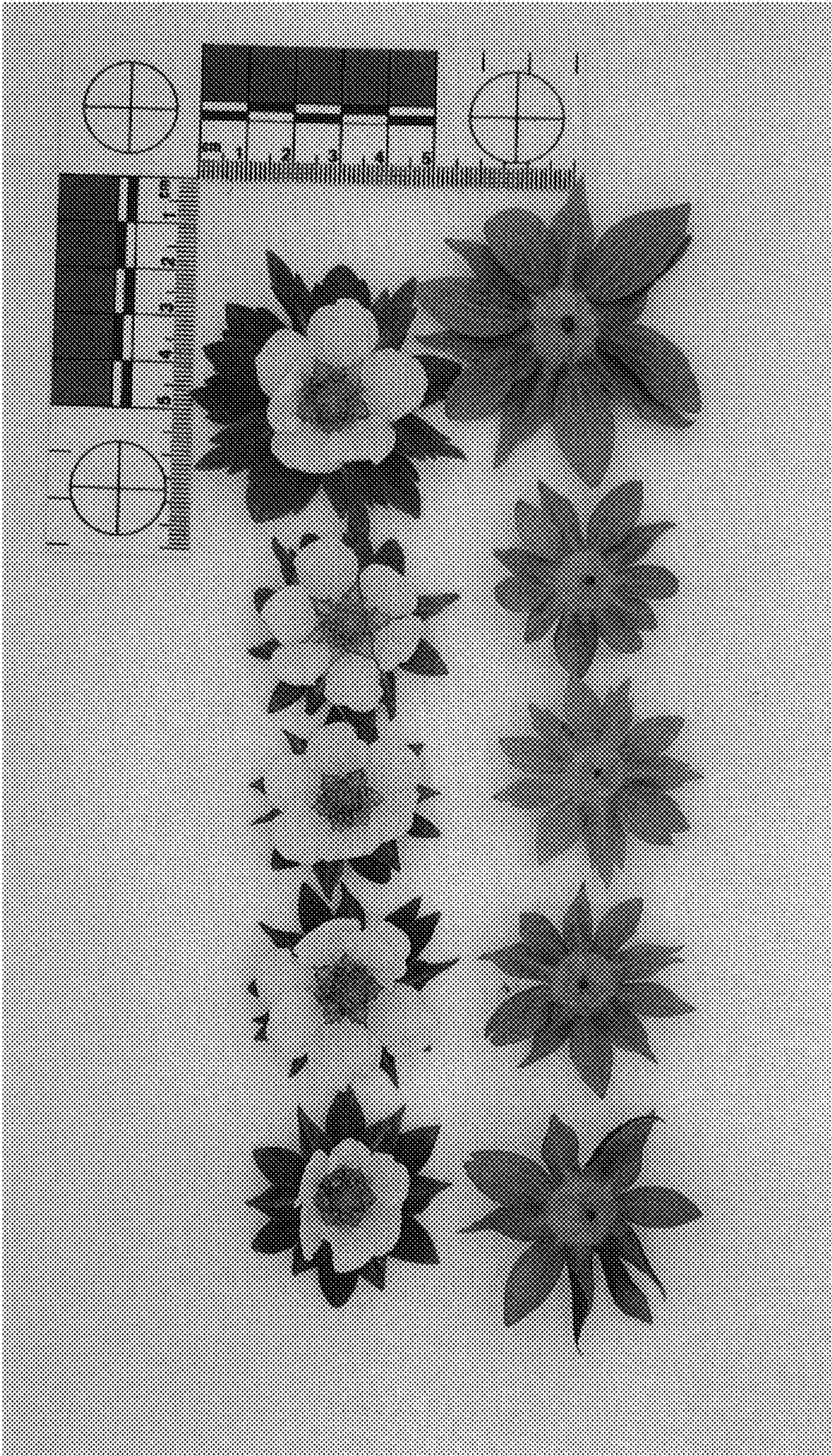


FIG. 3



FIG. 4



FIG. 5